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Using Deep Learning to Track Representational Flexibility Development of Children with Autism in a Virtual World

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***Abstract*—Representational Flexibility (RF) is an individual cognitive ability to select, coordinate and create alternative representations during information processing. Students with autism spectrum disorder (ASD) typically lack representational flexibility and therefore are underrepresented in the field of stem education. Research has used virtual reality-based (VR) 3D simulations to promote RF development in ASD adolescent learners. However, to promote RF development through VR, the need to be able to track RF throughout the sessions is critical. In this paper we describe the development of neural network classifiers, designed to track RF subsets throughout the session. We trained these classifies using different modes of data collected from 178 session recordings with eight ASD participants. Our empirical results show the promise and potential of using deep learning to provide real-time tracking of the cognitive-affective states of students with ASD. To the best of our knowledge, this is the first research effort to combine VR-training, educational data mining and deep learning to provide a nurturing and adaptive learning environment for students with ASD.**

Keywords—virtual/augmented reality, machine learning, artificial neural networks, computers for special needs

I. Introduction

Students with autism spectrum disorder (ASD) are an underrepresented group in STEM education [1]. Individuals with ASD experience developmental delays that include cognitive and social skill challenges that cause difficulty in STEM education. Representational flexibility, which includes the facility to use multiple external representations of information and the ability to switch flexibly between them, is a critical index for the practices of computational thinking and problem solving in the STEM domains in general. However, the development and enactment of representational flexibility in cognitive processes is effortful and demanding for most learners, but particularly so for students with ASD. Prior research observed that compared to their non-autistic peers, students with autism and a high heterogeneity experienced difficulty in holistic processing, cognitive shifting, and categorical induction (e.g., encoding general similarity and thematic relations). Yet cognitive training that aims to promote RF among heterogeneous learners with special needs is limited. Recent research has introduced virtual reality (VR)-based training to foster ASD students' RF development [2], [3]. VR enables students to perform various 3D design tasks, which can foster RF practices. Given that students with ASD's cognitive-affective states are dynamic, an essential component of VR-based training is the ability to track the RF state of learners to provide them with timely and adaptive learning supports. However, there is a lack of empirical research tracking ASD students' RF development during VR-based training. This study is part of a longitudinal project aimed at designing and studying individualized training of RF for computational and scientific problem solving among adolescents with ASD, in a VR based adaptive learning program called Force and Motion – Adaptive Representation (FM-AR). To rapidly and adaptively address the learners' needs, we aim to develop prediction models that track students' RF development using neural network driven classifiers, trained on the different modes of data that can be extracted from the VR Sessions.

II. Machine Learning using Educational Data

Different machine learning techniques have been applied in educational data to serve different purposes, such as predicting students' academic performance or persistence, developing adaptive learning systems, providing intelligent mentoring or instructional support, and automatically testing or grading students [4]-[7]. These machine learning techniques, which typically include clustering, classification, text mining, and social network analysis, are used for extracting salient relationships (or knowledge discovery) from the educational dataset. Only recently has the use of deep learning techniques emerged for data mining in educational environments [8],[9]. Particularly, deep learning techniques have been recently used to address a challenge that has gained increasing attention in mining educational data – knowledge tracing, or predicting students' knowledge or skill development based on their learning activity [10]. For example, Sharada et al. modeled student knowledge retention using deep learning and random forests with a dataset collected from a web-based mathematics tutor [11]. Okubo et al. used a neural network approach for students' performance prediction based on the logged learning activity data of students who used a learning management

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system, an e-portfolio system, and an e-book system [12]. They reported that deep learning (DL) outperformed the traditional machine learning baseline. Kim et al. proposed and used a DL algorithm, called GirtNet, to provide real-time student performance prediction [13]. Their results also showed that the proposed DL technique outperformed the chosen baseline. However, research is lacking on using deep learning for latent knowledge estimation where a student's knowledge is measured during online learning. This is distinct from the type of educational measurement common within tests in that during online learning, the knowledge is changing while it is being measured [14]. The few studies examining the DL technique for latent knowledge estimation reported mixed results governing the performance of DL models in comparison with traditional machine learning techniques [15]. It was reported that DL better exploits temporal information but tends to overfit a small dataset [16].

III. Integrated STEM Education in VR

Integrated STEM education requires students to perform information retrieval and processing in multifaceted learning tasks. Students need to gather and analyze data to understand the characteristics of a design challenge and its solution, as well as compare multiple representations of the solution through iterative testing and prototype revisions. RF is associated with heuristics that create solutions and strategies in problem-solving [17]. There is a growing interest in RF development in integrated STEM education. Research suggests that RF is related to general STEM competences [18]. The struggle of ASD students in integrated STEM education has been attributed to their low RF [2]. As demonstrated by ASD individuals' dynamic cognitive-affective states [19], learning-related stress may eventually lead to low levels of inhibition and attention switching [20]. ASD adolescents may encounter overwhelming obstacles when confronted with design quests, including complex information characteristics, which may lead to distraction and disengagement. Such difficulties in STEM education underline the importance of seeking a medium to foster RF practices in students with ASD. Using VR to perform inquiry-based, design problem-solving tasks fosters a deep understanding of integral STEM concepts [21]. Research also argues that RF includes the understanding of scientific concepts via various means (e.g., verbal description, graphs, and sketches). VR helps students develop the practices of forming different representations.

IV. Force and Motion – Adaptive Representation Template

An earlier study of the Authors explored whether and how VR-based design problem solving tasks aided in developing ASD children's perspective-taking skills and RF by supporting design-oriented interpersonal interactions [2]. Building on this previous work, we designed a VR and inquiry-based cognitive skills training program that engages ASD learners in a range of 3D-represented engineering design quests. Utilizing personalized learning and constructionism, FM-AR trains participants how to design and program motion-based simulations, ranging from interactive objects to computerized characters, in an open-source, 3D collaborative virtual world. FM-AR is aimed to facilitate participants' RF, or their ability to select, connect, convert, and construct multimodal representations of physics problems, while practicing computational concepts (e.g., sequencing, iteration, events, and conditionals). FM-AR offers a variety of virtual interactive worlds with specific design tasks. In this paper we will discuss the data extracted from two specific design quests or modules, Bridge Construction and NPC Design. In the Bridge Constructions module, the learners are tasked with designing and constructing a moveable bridge that allows for the progression of both land and the water traffic on the river that separates the two land masses. The design quest embeds the concepts related to Newtonian physics and the learners need to apply the presented information on the related concepts from physics and computer programming to complete the task. In the NPC Design Module, the participants use computer programming and logic to design their own non-player characters.



Fig. 1. FM-AR Training Module Examples (Left: Bridge design, Right: NPC design).

A critical part of the design and evaluation of the FM-AR program is evidence-centered and nonintrusive assessment of the practice or enactment of RF by the participants during FM-AR-supported learning activities. For that purpose, we have

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archived and collected the logged computational behaviors and verbal protocols (voice and text chats) of the study participants during their FM-AR sessions. Based on a recent critical review of the literature, we defined four subset skills that reflect RF that we aim to track during these VR-sessions [2], [3]. Attention-switching/cognitive shifting (ACS) refers to the process of switching attention or action based on changing rules and contextual demands. Alternative representation (AR) refers to an individual's skill in understanding and applying multiple representations. Pattern development (PD) is the ability to identify and delineate a pattern (or rules) during design problem-solving analysis. Finally, pattern contextualization (PC) refers to the acts of identifying and implementing the context of the endorsed pattern and/or customizing that pattern.

V. Experimental Data Acquistion and Processing

A. Data

We collected 178 audio and screen recordings of 1–1.5 hour VR training sessions with eight ASD adolescents. Additionally we collected the logged computational behaviors for each session. From all of these recordings we generated a transcript of the verbal protocols of all the participants' throughout their sessions. The resulting text was discretized into five-minute intervals. The corresponding audio and behavioral logs were similarly processed. The behavioral logs were transformed to quantify the frequency by which each behavioral feature occurs within each 5 min interval while an audio sentiment analysis quantified the audio logs within each interval. Together they formed the tabular section of our dataset. The resulting dataset contained 2203 data instances of text and tabular data.

B. Expert Labeling

Using both speech and log-data-mining techniques with supervised machine learning [2], we tracked and collected study participants' positive and negative enactments of the RF subsets throughout the VR sessions. For instance, in terms of ACS, once a study participant attended to multiple design tasks or items as needed, we coded it as a positive enactment of ACS. In other words, we assigned a positive label in ACS_P for all VR session intervals during which these types of behaviors took place. Similarly, if the study participant showed evidence of failing to switch attention between design tasks or items, we labeled this as a negative enactment of ACS i.e. we assigned a positive label in the ACS_N label for those time intervals. This expert labeling produced 8 labels overall, with 2 labels independently tracking positive and negative enactments of each of the four RF subsets. The gathered RF data was iteratively reviewed and corroborated by two expert coders. The coders independently reviewed the occurrences of RF subsets, then compared and discussed until there was 100% interrater agreement.

C. Classifiers

Multiple Neural Network classifiers were developed for each type of data gathered, with the goal of being able to independently label each enactment of the four RF subsets using binary classification, given a particular set of features. A multi-layered perceptron (MLP) made out of multiple dense layers was developed for RF classification using the tabular data. A neural network with a similar architecture was developed for classification using the text data, with an added initial pre-trained text embedding layer, a token based text embedding layer trained on English Google News [22]. Additionally, a classifier made out of a BERT preprocessor and encoder was also developed. Due to the apparent label imbalance in the labeled RF subset results, shown on Fig. 2, all classifiers were trained with random batches (RB) as well as a balanced batches (BB) i.e., batches with an equal number of negative and positive labels [23]. In the construction of the balanced batches for the classification of a specific label, a batch generator constructed training batches by pairing a unique set of sequential negatively label instances with an equal amount of randomly selected instances of with a positive label assignment. The batch size used for all classifiers was 32, and each classifier was trained for 20 epochs. Finally, for the purposes of this paper we focused on the positive enactment of the RF subsets, specifically, the labels ASC_P, AR_P, PD_P, PC_P. The performance of these classifiers was evaluated using the Accuracy, the AUC score and the distribution of correctly and incorrectly labeled predictions.

VI. Results & Discussion

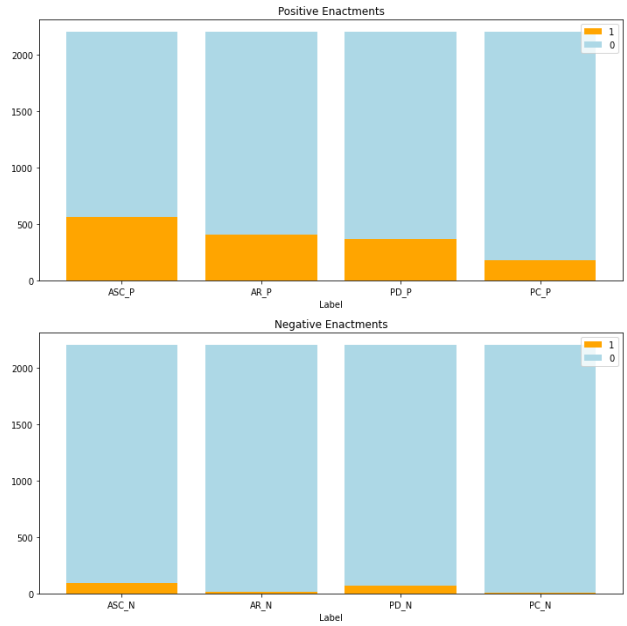


Fig. 2. Count of positive (1) and negative (0) instances for each label.

The results of the MLP classifier and classifier with the token-based text embedding layer (TTE), utilizing both types of batches, are presented on Fig. 3 and Fig. 4 respectively. In addition, an overview of each classifier's performance is presented on Fig. 5. The MLP featured had an architecture of only one dense intermediate layer. Additionally, in terms of the considered metrics, the BERT based classifier did not perform nearly as well as the other two models designed and therefore it is not featured in paper. From Fig. 5 we can observe that the MLP was able to perform better. This is further corroborated from the results displayed on Fig. 3 and 4.

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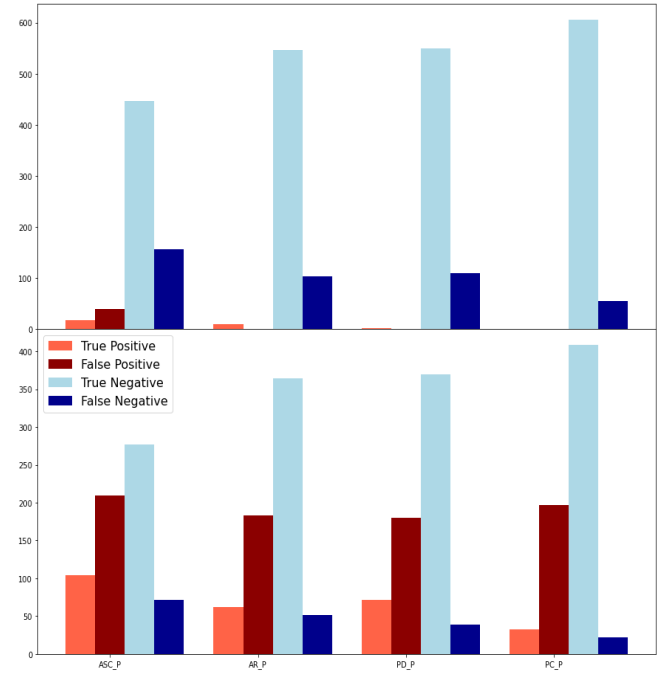


Fig. 3. Results of the MLP classifier using RB (top) and BB (bottom).

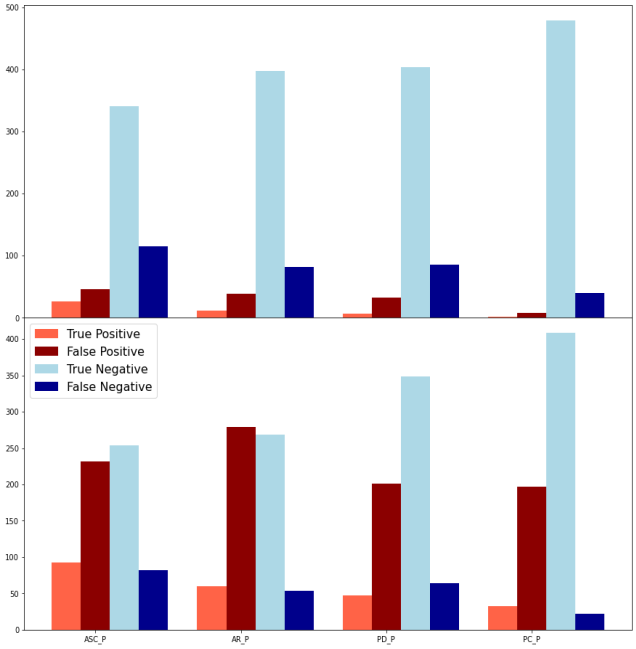


Fig. 4. Result of the TTE classifier using RB (top) and BB (bottom).

Despite the higher accuracy in classification using RB, Fig. 3 and 4 demonstrate a presence of negative label bias in this method of classifications. With the implementation of balanced batches with a preset ratio of positive to negative classes, there is better distribution of correctly labeled predictions i.e., an increase in true positives. However, this comes at cost to the accuracy of the classification, due to the classifier now being as likely to mislabel both classes, resulting in a significant amount of falsely labeled predictions. This could be attributed to the limited size of our dataset.

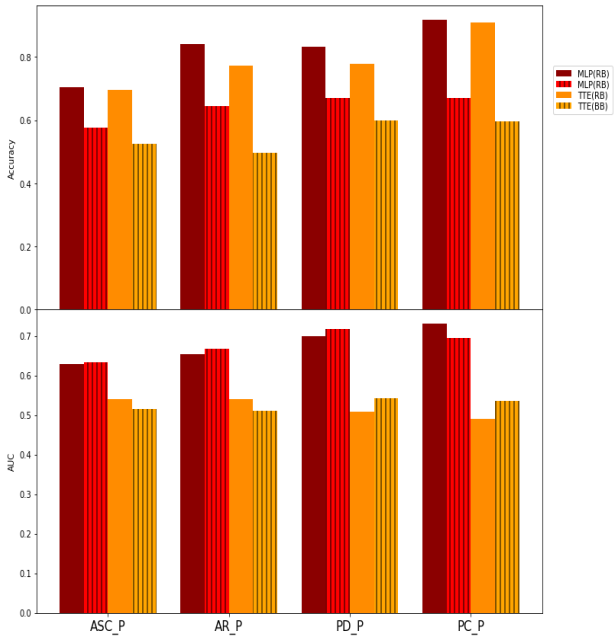


Fig. 5. Classifier performance.

VII. Conclusion

In this paper we present our work on developing neural networks classifiers with which we aim to track RF in students with ASD during VR training sessions. As the goal of the VR session is to promote positive RF development in students with ASD, the ability to track their dynamic cognitive-affective states is essential. The goal of these classifiers is to be able to rapidly and accurately track the outlined RF subsets in order for the VR-environment to provide adaptive support to counter any negative changes and foster proper growth. Our presented work is just the initial stepping stone of a novel approach of latent knowledge estimation. We have developed an adjustable VR based learning program for students with ASD that we hope to make automatically adaptive to students' cognitive-affective states with the result of producing a nurturing learning environment.

A. Limitations and Future Work

To address the issue of the small size of our dataset, we aim to increase the modes and volume of information that can be extracted from each session. Additionally, we plan to perform multimodal feature-level data fusion on the different modes of data extracted from the VR sessions.

References

[1] Ehsan, H., Eispoli, M., Lory, C., and Gregori, E, "A systematic review of STEM instruction with students with autism spectrum disorders" *Review Journal of Autism and Developmental Disorders*, 5, 4, 327–348, 2018.

[2] Moon, J., Ke, F., & Sokolikj, Z., "Automatic assessment of cognitive and emotional states in virtual reality-based flexibility training for four adolescents with autism", *British Journal of Educational Technology*, 51(5), 1766-1784, 2020.

[3] Ke, F., Moon, J., & Sokolikj, Z., "Virtual reality-based social skills training for children with autism spectrum disorder", *Journal of Special Education Technology*, 37(1), 49-62, 2022

Authorized licensed use limited to: University of New South Wales. Downloaded on September 21,2025 at 16:11:21 UTC from IEEE Xplore. Restrictions apply.

[4] Baker, R. S., Martin, T., & Rossi, L. M., "Educational data mining and learning analytics", *The Wiley handbook of cognition and assessment: Frameworks, methodologies, and applications*, 379-396, 2016.

[5] Coelho, O. B., & Silveira, I., "Deep learning applied to learning analytics and educational data mining: A systematic literature review", *In Brazilian Symposium on Computers in Education (Simpósio Brasileiro de Informát-ica na Educação-SBIE)* (Vol. 28, No. 1, p. 143), 2017.

[6] Khan, A., & Ghosh, S. K., "Student performance analysis and prediction in classroom learning: A review of educational data mining studies", *Education and Information Technologies*, 26(1), 205-240, 2021.

[7] Salloum, S. A., Alshurideh, M., Elnagar, A., & Shaalan, K., "Mining in educational data: review and fu-ture directions", *In The International Conference on Artifi-cial Intelligence and Computer Vision* (pp. 92-102). Springer, Cham, 2020.

[8] Doleck, T., Lemay, D. J., Basnet, R. B., & Bazelais, P., "Predictive analytics in education: a comparison of deep learning frameworks", *Education and Information Technologies*, 25(3), 1951-1963, 2020.

[9] Hernández-Blanco, A., Herrera-Flores, B., Tomás, D., & Navarro-Colorado, B., "A systematic review of deep learning approaches to educational data mining", *Complexity*, 2019.

[10] Piech, C., Bassen, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L. J., & Sohl-Dickstein, J., "Deep knowledge tracing", *Advances in neural information processing systems*, 28, 2015

[11] N. Sharada, M. Shashi, and X. Xiong, "Modeling student knowledge retention using deep learning and random for-ests," *Journal of Engineering and Applied Sciences*, vol. 13, no. 6, pp.1347–1353, 2018.

[12] F. Okubo, T. Yamashita, A. Shimada, and H. Ogata, "A neural network approach for students' performance prediction," *Proceedings of the the Seventh International Learning Analytics& Knowledge Conference* (LAK '17), pp. 598-599, ACM., New York, NY, USA, March 2017.

[13] Kim, B. H., Vizitei, E., & Ganapathi, V., "GritNet: Student performance prediction with deep learning", arXiv preprint arXiv:1804.07405, 2018.

[14] Slater, S., Joksimović, S., Kovanovic, V., Baker, R. S., & Gasevic, D., "Tools for educational data mining: A review", *Journal of Educational and Behavioral Statistics*, 42(1), 85-106, 2017.

[15] Wang, L., Sy, A., Liu, L., & Piech, C., "Learning to represent student knowledge on programming exercises using deep learning", *Proceedings of the 10th International Educational Data Mining Society*, 324-329, 2017.

[16] Gervet, T., Koedinger, K., Schneider, J., & Mitchell, T., "When is deep learning the best approach to knowledge tracing?", *Journal of Educational Data Mining*, 12(3), 31-54, 2020.

[17] Nistal, A. A., Van Dooren, W., Clarebout, G., Elen, J., and Verschaffel, L., "Conceptualising, investigating and stimulating representational flexibility in mathematical problem solving and learning: a critical review", *ZDM*, 41, 5, 627–636, 2009.

[18] Pande, P., and Chandrasekharan, S., "Representational competence: towards a distributed and embodied cognition account", *Studies in Science Education*, 53, 1, 1–43, 2017.

[19] Messinger, D. S., Duvivier, L. L., Warren, Z. E., Mahoor, M., Baker, J., Warlaumont, A., and Ruvolo, P., Affective computing, emotional development, and autism, 2015

[20] De Vries, M., and Geurts, H., "Influence of autism traits and executive functioning on quality of life in children with an autism spectrum disorder", *Journal of Autism and Developmental Disorders*, 45, 9, 2734–2743, 2015.

[21] Makransky, G., and Petersen, G. B., "The Cognitive Affective Model of Immersive Learning (CAMIL): a Theoretical Research-Based Model of Learning in Immersive Virtual Reality," *Educational Psychology Review*, 1–22, 2021.

[22] Y. Bengio, R. Ducharme, P. Vincent, C. Jauvin. "A Neural Probabilistic Language Model," *Journal of Machine Learn-ing Research*, 3:1137-1155, 2003.

[23] Y. Yan, M. Chen, M. -L. Shyu and S. -C. Chen, "Deep Learning for Imbalanced Multimedia Data Classification," 2015 IEEE International Symposium on Multimedia (ISM), 2015, pp. 483-488, doi: 10.1109/ISM.2015.126.

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